

Room-01 → Lab-04 Route Map — Department Building

Route-finding problem modeled as a graph and solved using Breadth-First Search (BFS)

1. Problem Statement

Starting point: Room-01 on the Ground Floor. Goal: Lab-04 on the First Floor.

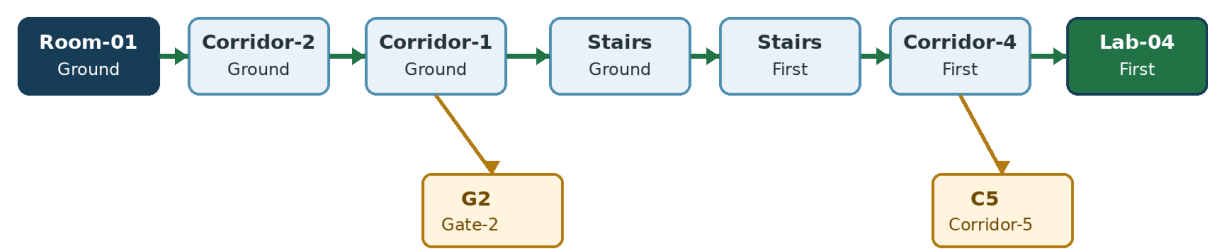
For this graph model, Room-01 connects to Corridor-2, which leads to Corridor-1 and the ground-floor staircase. After moving upstairs, the route continues through Corridor-4 to Lab-04. Gate-2 and Corridor-5 are treated as nearby side branches visible in the sketch, but they are not required for the shortest route.

2. Route Map (Locations Coded)

Code	Location	Floor
R1	Room-01	Ground
C2	Corridor-2	Ground
C1	Corridor-1	Ground
G2	Gate-2	Ground
ST-G	Stairs (ground landing)	Ground
ST-1	Stairs (first-floor landing)	First
C4	Corridor-4	First
C5	Corridor-5	First
L4	Lab-04	First

Figure 1: Simplified route network from Room-01 (Ground Floor) to Lab-04 (First Floor).

Room-01 to Lab-04 — Coded Route Network



Green arrows show the BFS shortest route. Gold branches are valid nearby locations but are not needed to reach Lab-04.

3. Graph Representation (Adjacency List)

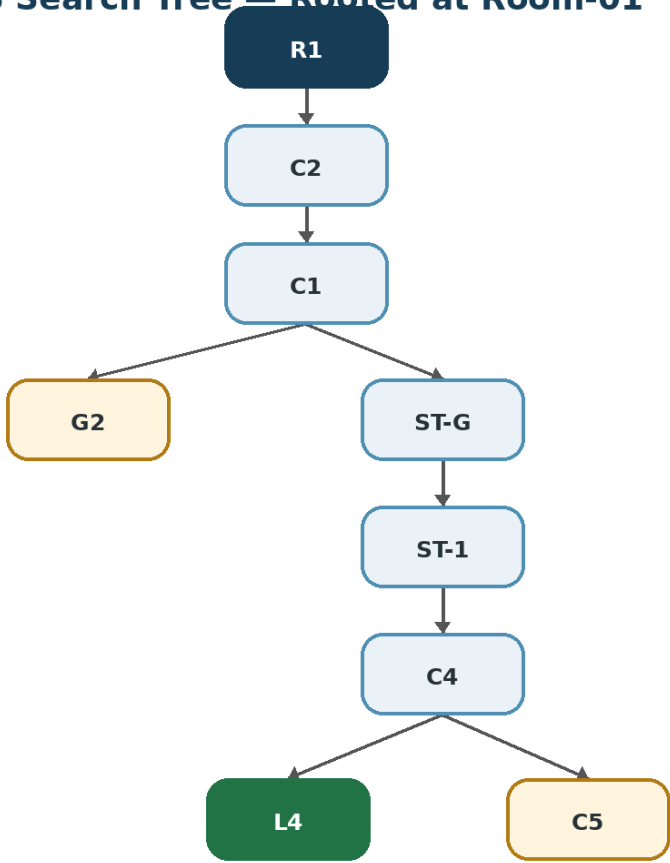
Following a State → Action → New State representation, the selected part of the building layout is converted into a graph. Each node is a location and each edge is a direct walkable connection.

Node	Connects to
R1 (Room-01)	C2
C2 (Corridor-2)	R1, C1
C1 (Corridor-1)	C2, G2, ST-G
G2 (Gate-2)	C1
ST-G (Ground stairs)	C1, ST-1
ST-1 (First-floor stairs)	ST-G, C4
C4 (Corridor-4)	ST-1, L4, C5
C5 (Corridor-5)	C4
L4 (Lab-04)	C4

4. BFS Search Tree

The search tree is rooted at Room-01 . BFS expands the building graph level by level. Gate-2 is explored as a side branch from Corridor-1. On the first floor, Lab-04 and Corridor-5 are both reachable from Corridor-4, but Lab-04 is the goal state.

BFS Search Tree – Rooted at Room-01



Goal node Lab-04 is reached at depth 6. Gate-2 and Corridor-5 are side branches.

Figure 2: BFS search tree rooted at Room-01.

5. BFS Trace (Step by Step)

BFS(problem): start at R1, enqueue it, repeatedly dequeue the front node, test it against the goal (L4), and enqueue any unvisited neighbours.

Step 0: Initialization

Queue: [R1]
Visited: {R1}

Step 1: Dequeue R1 → Expand (C2)

Queue: [C2]
Visited: {R1, C2}

Step 2: Dequeue C2 → Expand (R1, C1)

R1 already visited [X] Add C1 [OK]
Queue: [C1]
Visited: {R1, C2, C1}

Step 3: Dequeue C1 → Expand (C2, G2, ST-G)

C2 already visited [X] Add G2 [OK] Add ST-G [OK]
Queue: [G2, ST-G]
Visited: {..., G2, ST-G}

Step 4: Dequeue G2 → Expand (C1)

C1 already visited [X]
Queue: [ST-G]

Step 5: Dequeue ST-G → Expand (C1, ST-1)

C1 already visited [X] Add ST-1 [OK]
Queue: [ST-1]
Visited: {..., ST-1}

Step 6: Dequeue ST-1 → Expand (ST-G, C4)

ST-G already visited [X] Add C4 [OK]
Queue: [C4]
Visited: {..., C4}

Step 7: Dequeue C4 → Expand (ST-1, L4, C5)

ST-1 already visited [X] Add L4 [OK] Add C5 [OK]
Queue: [L4, C5]
Visited: {..., L4, C5} ← destination discovered

Step 8: Dequeue L4 → GOAL FOUND!

Search stops because Lab-04 is the destination.

BFS Order of Expansion: R1 → C2 → C1 → G2 → ST-G → ST-1 → C4 → L4

Node	Parent
C2	R1
C1	C2
G2	C1
ST-G	C1
ST-1	ST-G
C4	ST-1
L4	C4
C5	C4

6. Shortest Route Found by BFS

Tracing the parent links backward from Lab-04 gives the shortest unweighted path in the selected graph:

R1 → C2 → C1 → ST-G → ST-1 → C4 → L4 (6 edges / steps)

In location names: Room-01 → Corridor-2 → Corridor-1 → Ground-Floor Stairs → First-Floor Stairs → Corridor-4 → Lab-04.

BFS selects this route because it reaches the goal using the minimum number of edges. Gate-2 and Corridor-5 are side branches and do not reduce the number of steps required to reach Lab-04.

7. State-Space Summary

Element	Description
Initial State	R1 — Room-01, Ground Floor
Goal State	L4 — Lab-04, First Floor
Actions	Move through an adjacent corridor/passage or use the staircase
Path Cost	Each graph edge is treated as one equal-cost step
Search Strategy	Breadth-First Search (BFS)
Result	Shortest route contains 6 edges

8. Original Hand-Drawn Floor Plan (Provided Reference)

The image below is the provided building sketch used to select and model the Room-01 to Lab-04 route. It has been rotated only for easier reading.

